Power BI KPI Creation, Formatting, Quick Measures & Filtering

Creating KPIs (Key Performance Indicators) in Power BI is one of the most common and essential tasks in developing any analytical dashboard. KPIs summarize key metrics in a clear, visually appealing, and interactive manner. This document explains, in detail, how various KPIs were created, formatted, duplicated, optimized, and filtered as per the business requirement. It also covers the concepts of **New Measure**, **Quick Measure**, **Format Painter**, and **Visual-Level Filters**, along with layout optimization techniques for dashboard design.

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1. Creating KPI Cards in Power BI

KPI visuals in Power BI allow you to display an aggregated metric such as total sales, total orders, total profit, or profitability.
The process typically involves the following steps:

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1.1 Adding a KPI Card

1. Navigate to the **Report View**.
2. From the **Visualizations** pane, select the **Card** visual.
3. Drag an appropriate numerical field (e.g., *Total Sales*, *Quantity*, *Profit*) into the **Fields** area.

Immediately, Power BI calculates the aggregate (usually SUM by default) and shows it inside the card.

1.2 Renaming the Card Title

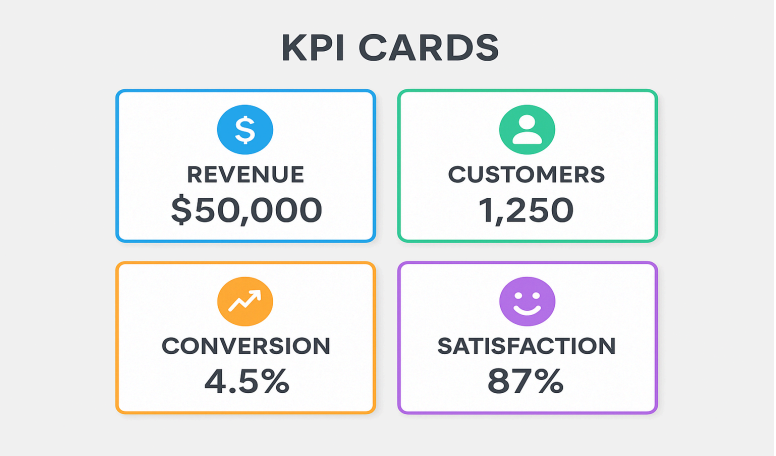
To ensure clarity, every card should have a descriptive title:

1. Select the card.
2. Go to **Format Visual > Title**.
3. Turn the title ON (if not already).
4. Enter a meaningful name such as *Total Orders*, *Total Sales*, *Total Profit*, etc.

Consistent naming improves readability and guides the user logically across the report.

2. Formatting KPI Cards

Visual consistency across all KPIs greatly improves dashboard quality. Formatting includes adjusting the sizes of:



- **Callout Value** (the large number displayed)
- **Title Text**
- Card size, width, and alignment
- Font size, font color, background, borders, etc.

2.1 Adjusting Callout Value Size

To modify the numerical value display:

1. Select the KPI card.
2. Go to **Format Visual → Callout Value**.
3. Change **Text size** (e.g., set to 25 or 30 depending on your layout).

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This is done to ensure that the metric is readable but not oversized.

2.2 Adjusting Title Size

Similarly:

1. Go to **Title → Text Size**.
2. Adjust title size to match the callout value or design needs.

A common mistake is keeping large fonts across all visuals, which consumes excessive space. Optimal font sizes help fit all KPIs neatly in a single row.

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3. Duplicating KPI Cards (Copy + Paste)

Instead of creating each KPI from scratch, Power BI allows duplicating an existing card:

- Select any existing card.
- Press **Ctrl + C** (Copy).
- Press **Ctrl + V** (Paste).

This creates an exact clone with:

- the same formatting,
- same width & height,
- same font size, colors, and background.

You only need to:

- replace the metric,
- update the title.

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Duplicating cards saves considerable time during dashboard development.

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4. Using Format Painter for Consistent Styling

Power BI includes a **Format Painter**, similar to Microsoft Office applications.

4.1 Purpose of Format Painter

It ensures **uniform formatting** across multiple visuals.

4.2 How to Use It

1. Select the card with the desired styling.
2. Click the **Format Painter** icon (a paintbrush) in the Home ribbon.
3. Click on any other card visual.

Immediately, the second card will adopt:

- the same text sizes,
- same background,
- same title style,
- same callout value style,
- same border/shadow settings.

This ensures your dashboard looks professional and consistent.

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4.3 Applying to All KPIs

The steps are repeated for all KPI cards to make the entire row visually uniform.

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5. Creating Measures Using “New Measure”

Power BI uses DAX (Data Analysis Expressions) to create custom calculations.
The **New Measure** option is used when you want full control over the formula.

5.1 Example: Profitability Calculation

Profitability is calculated as:

```
Profitability = SUM(Profit) / SUM(Sales)
```

This measure represents the profit margin and is displayed as a percentage.

5.2 Steps to Create a New Measure

1. Select the relevant table.
2. Go to **Modeling → New Measure**.
3. Write the DAX formula manually.
4. Press Enter.
5. Format the measure as a percent in the **Measure Tools** pane.

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6. Creating Measures Using "Quick Measure"

Power BI provides **Quick Measures**, which generate DAX formulas automatically based on predefined templates.

6.1 When to Use Quick Measure

Use it when:

- the calculation is standard (e.g., division, multiplication, filters),
- you need quick results without writing DAX manually.

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6.2 Steps to Create a Quick Measure

1. Select a table.
2. Go to **Modeling → Quick Measure**.
3. In the Quick Measure window:
    - Choose a category (e.g., **Mathematical Operations**).
    - Select **Division**.
4. Provide:
    - **Numerator** → drag *Profit* field.
    - **Denominator** → drag *Sales* field.

5. Click **Add**.

Power BI automatically generates a formula like:

```
Profitability_New = DIVIDE(SUM(Profit), SUM(Sales))
```

6.3 Comparing New Measure vs Quick Measure

Both methods produce the same final result but serve different purposes:

Feature	New Measure	Quick Measure
Write DAX manually	Yes	No
Ideal for complex logic	Yes	Limited
Ideal for beginners	No	Yes
Control over formula	Full	Limited

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6.4 Removing Duplicate Measures

Since two profitability measures were created, one was deleted:

- Click the **three dots** next to the measure.
- Select **Delete from model**.

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7. Creating KPI for "Quantity for Standard Shipping & Accessories"

The requirement:

> Display the quantity only for orders where
> **Shipping Mode = Standard Class**
> **Subcategory = Accessories**

This requires:

1. Creating a duplicate of the Total Quantity card.
2. Applying **visual-level filters**.

7.1 Duplicating the Total Quantity Card



- Copy the existing quantity card.
- Paste it.
- Change the title to something meaningful, such as:

“Quantity – Standard Accessories”

7.2 Applying Filters

Power BI enables filtering visuals at 3 levels:

- Visual level
- Page level
- Report level

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We use **Visual-level filters** here.

Steps:

7.2.1 Filter by Ship Mode

1. Drag **Ship Mode** field into the Filters pane (visual-level).
2. Use **Basic Filtering**.
3. Check **Standard Class**.

Now the quantity changes automatically (from 12,000 to ~7,142 in your example).

7.2.2 Filter by Subcategory

1. Drag **Subcategory** field into Filters (for the same card).
2. Select **Accessories**.

The value updates again (to ~6,022 in your example).

This demonstrates that multiple filters can be applied simultaneously to the same KPI visual.

8. Understanding Filters and How They Work

8.1 Basic Filtering

Lists all possible values, allowing checkboxes.

8.2 Advanced Filtering

Allows conditions such as:

- "contains"
- "greater than"
- "starts with"
- "is not blank"

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8.3 Relative Date Filtering

Used frequently for date-based dashboards.

8.4 Multiple Filters Combined

Each filter condition is applied using **logical AND**, meaning:

- Both conditions must be true for a record to be included.

This is how we isolated:

Standard Shipping + Accessories Subcategory

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9. Optimizing Layout and Sizing of KPI Cards

A professional dashboard places multiple KPIs in a single horizontal row for maximum readability. To achieve this, you may need to:

- Reduce font sizes
- Reduce card widths
- Remove unnecessary white space
- Align cards uniformly

9.1 Reducing Font Size

Example:

- Change callout value from **25 → 20**
- Change title from **25 → 20**
- Ensure consistency across all cards

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9.2 Adjusting Card Size

Use blue handles on visuals to adjust width and height.

9.3 Using Gridlines / Snap-to-grid

Ensures perfect alignment.

9.4 Moving Cards Neatly

Use arrow keys for fine adjustment.

9.5 Reducing Logo Size

To free up more horizontal space.

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10. Performance Considerations

While adjusting visuals, Power BI may process queries in the background. This can temporarily slow down the interface.

Tips:

- Avoid clicking repeatedly when the interface is loading.
- Wait for the visual to update.
- Use a laptop with good specifications for a smoother experience.

11. Final Review of KPI Layout

After applying formatting, filters, and optimizing layout:

- All KPIs were placed in a **single row**.
- Cards were resized uniformly using **Format Painter**.
- Titles and values were made consistent.
- Filters were applied correctly for the “Standard Accessories Quantity” KPI.
- Duplicate profitability measure was deleted.
- The final dashboard section looks clean, compact, and professional.

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12. Summary of Key Concepts Covered

12.1 KPI Card Creation

- Adding card visuals
- Assigning fields
- Formatting titles and values

12.2 Formatting Tools

- Font sizing
- Adjusting card layout
- Format Painter

12.3 Measures

- Creating custom DAX measures
- Using Quick Measures
- DIVIDE function vs simple arithmetic
- Deleting unnecessary measures

12.4 Filtering

- Visual-level filters
- Applying multiple filters (AND conditions)
- Using categorical selections

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12.5 Layout Optimization

- Arranging KPIs in a single row
- Reducing visual footprint
- Proper alignment
- Adjusting logo size

12.6 Performance Tips

- Avoid multiple clicks during query refresh
- Save file periodically

13.Conclusion

Building KPIs in Power BI involves more than just placing card visuals—it requires attention to detail in formatting, filtering, layout design, consistency, and measure creation. Understanding both **New Measures** and **Quick Measures** empowers you to implement metrics in the most efficient way depending on your skill level and project requirements. Using tools like **Format Painter** speeds up the development process while ensuring a professional appearance. Applying filters correctly allows targeted KPIs to reflect precise business logic, such as filtering for a specific shipping mode and subcategory.

By mastering these techniques, you can create clean, accurate, interactive, and highly insightful KPI dashboards that communicate business performance effectively.

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